

International Effects of Stock Market Dispersion

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Abstract

We study the extent to which stock market dispersion is related to unemployment and output growth for 16 countries over 20 years. Using panel VARs and panel dynamic regressions, we find increases in stock market dispersion across industries to induce future increases in unemployment and future decreases in industrial production (IP) growth. Moreover, the responses of unemployment and IP growth following a positive shock to stock market dispersion are persistent and are robust to various controls, sample periods, and estimation methods. Our paper provides cross-country evidence in support of the hypothesis that shifts in employment demand across industries negatively affect employment.

Keywords: Stock Market Dispersion, Panel VARs, Sector Shifts

JEL Classification: E4; E6; C23

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1 Introduction

In the presence of labor market frictions, shifts in demand across sectors are costly for both the employees and the firms, resulting in displacements of resources that might not be immediately offset by more productive industries. Since stock market returns are indicative of investors' expectation of future profits, these displacements are reflected in the standard deviation of returns across industries (i.e., stock market dispersion). In this sense, an increase in stock market dispersion is expected to predict an increase in unemployment, or subdued economic performance in general. While evidence of this sector shifts hypothesis has been well documented in earlier single-country investigations, whether such evidence holds when taking into account cross-country dynamics is less clear.¹

In this paper, we provide international evidence on how the dispersion of stock market returns across industries, as a proxy for sector shifts, is able to explain short-run movements in the unemployment rate and output growth. Our empirical approach revolves around the use of panel vector-auto-regressions (VARs) to estimate the effects of stock market dispersion on the labor market (via the unemployment rate) and on the economy in general (via industrial production (IP) growth) while also accounting for unobserved interdependencies across countries.² This approach is especially relevant, given the co-movement of international business cycles well-documented in the literature (see, e.g., [Crucini et al. \(2011\)](#) or [Ayhan Kose et al. \(2008\)](#)). One source of the novelty of the paper, therefore, is to examine the effects of stock market dispersion - as a proxy for sector shifts - on aggregate activity in a dynamic panel VAR setting, taking into account international linkages. Moreover, since the literature has largely been silent on the impact of stock market dispersion on output growth, our exploration of this relationship is a

¹See, e.g., [Lilien \(1982\)](#), [Loungani et al. \(1990\)](#), [Brainard and Cutler \(1993\)](#), [Loungani and Trehan \(1997\)](#), [Fortin and Araar \(1997\)](#), and more recently [Chodorow-Reich and Wieland \(forthcoming\)](#).

²As carefully documented by [Canova and Ciccarelli \(2013\)](#), one advantage of using panel VARs over single-country VARs is the ability to control for cross-country spillovers. These spillovers are well-documented via various channels in the literature, with examples including (1) the institution spillovers channel (e.g., [Ranjan \(2013\)](#), [Agrawal et al. \(2016\)](#), and [Heid and Larch \(2016\)](#)), (2) the technology spillovers channel (e.g., [Goel \(2017\)](#)), and (3) the offshoring spillovers channel (e.g., [Groizard et al. \(2014\)](#), [Hummels et al. \(2014\)](#), [Harrison and McMillan \(2011\)](#), and [Wright \(2014\)](#), among others).

natural extension to the literature, given that unemployment and output growth are typically correlated over the business cycle.³

Turning to more details of our empirical approach, we construct a panel of 16 developed countries from 1994M1 to 2017M5 to understand how unemployment and output growth depend on stock market dispersion across industries. In the first pass using dynamic panel regressions, we find the coefficients of the lag(s) of stock market dispersion on the unemployment rate and industrial production (IP) growth - as a proxy for output growth - to be statistically significant. Moreover, these coefficients are positive for the unemployment rate and negative for IP growth. We also find these effects to be particularly pronounced during the periods following the Great Recession.

We next explore the extent to which macroeconomic aggregates, specifically unemployment and IP growth, respond to an unexpected increase in stock market dispersion using panel VARs that control for both country-specific characteristics and time-varying shocks that affect all countries. Since macroeconomic shocks across countries are known to be driven by a common global dynamic factor in the literature (see, e.g., [Crucini et al. \(2011\)](#)), these controls are intended to account for the international linkages that are driven by this common factor.

Turning to the results, we find unemployment rises following a positive shock to stock market dispersion, while output, as measured by IP growth, responds negatively to the same shock. We also find these responses of unemployment and IP growth to be long-lived, lasting well beyond the 15th month after the initial shock. Overall, the results from both the dynamic panel regressions and the panel VARs are consistent when controlling for economic policy uncertainty, for implied stock market volatility, for consumer confidence, and for oil price volatility. Our results dovetail with the theoretical prediction of [Lilien \(1982\)](#) and [Davis \(1987\)](#) that sector shifts, as measured by increases in stock market dispersion, are negatively associated with employment and implicitly, with output growth.

³[Fortin and Araar \(1997\)](#) study the effects of stock market dispersion using Canadian data. Recently, [Angelidis et al. \(2015\)](#) find that returns dispersion can help predict variation in economic activity using a Probit regression with data from G-7 countries. While also exploring the panel nature of the data, their approach is different from our panel VAR approach that highlights both the cross-sectional heterogeneity and the dynamic nature of cross-country interdependencies.

In principle, stock market dispersion - as a proxy for sector shifts - can capture changes in both sector *structures* and the *levels* of aggregate demand since this measure is constructed from stock market returns. Empirically, this observational equivalence problem - first noted by [Abraham and Katz \(1986\)](#) and later by [Loungani et al. \(1990\)](#) - does not appear to be the case for stock market dispersion, given how little its correlation with unemployment is attributable to movements in aggregate demand (see, e.g., [Loungani et al. \(1990\)](#), and to some extent, [Neumann and Topel \(1991\)](#)).⁴ Moreover, since the correlation between stock market dispersion and unemployment remains significant with years of lags, [Loungani et al. \(1990\)](#) argue that the use of stock market dispersion, as opposed to the dispersion of employment growth across industries, does not necessarily suffer from the observational equivalence criticism by [Abraham and Katz \(1986\)](#).⁵ In our analysis, it is reassuring to find that the results are consistent regardless of whether controls for changes in the levels of aggregate demand are included.

The rest of this paper is organized as follows. Section 2 describes the data and details the construction of stock market dispersion. Section 3 presents the result using dynamic panel regressions that economic performance, measured by the unemployment rate and IP growth, is significantly dependent on the lags of stock market dispersion. Section 4 documents the extent to which the unemployment rate and IP growth respond to an unexpected increase in stock market dispersion in a panel VAR setting. We conclude in Section 5.

2 Data

Our dataset is a monthly panel of stock market dispersion and selected macroeconomic aggregates across 16 countries from 1994M1 to 2017M5. The set of 16 countries - chosen based on the availability of data - includes Austria, Belgium, Canada, Denmark, Finland, France, Germany,

⁴[Neumann and Topel \(1991\)](#) decompose industry-level unemployment shares into the structural and cyclical components to construct a measure of structural changes. Consistent with the sector shifts hypothesis, they find this measure to be able to significantly explain movements in unemployment.

⁵Using postwar U.S. data, [Lilien \(1982\)](#) documents a significant and positive correlation between the lags of the variance of employment growth across sectors and the aggregate unemployment rate. Despite considerable support in the literature (e.g., [Grossman et al. \(1983\)](#) and [Barro \(1984\)](#)), open criticism - most prominently by [Abraham and Katz \(1986\)](#) - remains that such a positive correlation does not necessarily arise solely from sector shifts.

Ireland, Italy, Japan, the Netherlands, Norway, Spain, Sweden, the United Kingdom, and the United States.

2.1 Data Sources and Variable Construction

Stock Market Dispersion We construct stock market dispersion from the daily industry-level Financial Times Stock Exchange (FTSE) stock indexes. As the FTSE indexes are not available for every industry across all 16 countries in the sample, we pick the 12 industries that are common across our set of countries. These industries include banking, electricity, finance, mass media, manufacturing, consumer goods, mining, technology, telecommunications, health care, oil and gas, and utilities. For each combination of country and industry, we normalize so that the values on Jan 01, 2010 are 100. Using these daily indexes, we construct the returns R_{ijt} for each industry j in country i in day t :

$$R_{ijt} = \frac{(D_{ijt} - D_{ijt-1})}{D_{ijt-1}}, \quad (1)$$

where D_{ijt} represents the daily FTSE value for industry j in country i . We next obtain the *weighted* standard deviation σ_{it}^R of daily stock market returns for each country i at time t across all industries as follows

$$\sigma_{it}^R = \sqrt{w_i \{R_{ijt} - \bar{R}_{it}\}^2}, \quad (2)$$

where w_i is the weight assigned for each industry in terms of their total equivalent shares of aggregate output. Our construction of stock market dispersion follows [Loungani et al. \(1990\)](#), who also employ a similarly weighted standard deviation of stock returns.⁶ To obtain monthly stock market dispersion, we average the daily values of σ_{it}^R over a month.

[Figure 1 Here]

To get a sense of how stock market dispersion varies over the business cycle, we plot its cross-country mean in Figure 1. At first glance, stock market dispersion is at its highest point

⁶Similar to [Loungani et al. \(1990\)](#), we also find that our results are consistent whether equal weights or the output-share weights w_i are used for the construction of σ_{it}^R .

during the Great Recession. While not drawing any conclusion solely from this observation, we find it consistent with the notion that periods of large dispersion in stock market returns across industries coincide with periods of subdued economic performance.

Other Variables Our dependent variables are the unemployment rate and IP growth. The unemployment rate is the harmonized unemployment rate series from the OECD Stats database and is first-differenced. Our IP growth data are monthly and are also from the OECD Stats database. Unless otherwise noted, all monthly raw variables are obtained seasonally adjusted before any transformation.

[Table 1 Here]

We obtain the Economic Policy Uncertainty (EPU) indices from [Baker et al. \(2016\)](#), for the U.S., Europe, and Japan, as controls. We also construct, and seasonally adjust at the monthly frequency, a measure of realized oil volatility by taking the standard deviation of the crude oil prices over a month using West Texas Intermediate (WTI) daily oil price from the St. Louis' FRED. The daily Chicago Board Options Exchange (CBOE) S&P 500 Volatility Index (VIX) (CBOE:VIX) index, averaged out at the monthly frequency, is used to control for uncertainty arising from the stock market. The remainders of our set of controls include: a measure of short-term (three-month) interest rate from the OECD Stats database, the Michigan Consumer Sentiment Index (MCI) and the Consumer Confidence Index (CCI) series from the OECD Stats database, the last of which measures households' plans for major purchases and their expectations for the future. Table 1 provides a list of countries in our sample, along with the means and standard deviations of stock market dispersion, IP growth, and the unemployment rate.

3 Dynamic Panel Regression

As a preamble to our panel VAR analysis, we first investigate the extent to which stock market dispersion is related to economic performance using dynamic panel regressions. We do so with

two measures of economic performance: (1) the unemployment rate and (2) IP growth. The key result here is that increases in stock market dispersion are associated with depressed economic performance, as evidenced by the negative coefficients on IP growth and the positive coefficients on the unemployment rate. This result is robust to the addition of various controls and to different sample periods.

3.1 Dynamic Panel Regression

We estimate the following dynamic panel regression

$$Y_{it} = \alpha + \sum_{k=1}^K \beta_k Y_{it-k} + \sum_{k=1}^K \gamma_k X_{it-k} + \varepsilon_{it} \quad (3)$$

in which Y_{it} represents either IP growth or the change in the unemployment rate for country i at time t and k here denotes the corresponding lags of dependent or independent variables. Depending on the exact specification, X_{it} includes stock market dispersion, and the following controls: the change in the interest rate, the change in MCI, the change in CCI, the change in EPU, the CBOE:VIX index, and a measure of oil volatility. To estimate the dynamic panel regression in Equation 3, we use a least square dummy variable (LSDV) panel regression with heteroscedasticity and auto-correlation consistent (HAC) standard errors.⁷ Since the LSDV estimator is also sometimes referred to as the fixed effects estimator, we shall refer to them interchangeably.⁸ As carefully documented by [Judson and Owen \(1999\)](#) and [Kiviet \(1995\)](#), an LSDV dynamic panel regression using lagged dependent variables performs relatively well when the time dimension is relatively large ($T > 30$).⁹ In particular, [Judson and Owen \(1999\)](#) find that dynamic panel regressions in the spirit of Equation 3 are asymptotically valid when the time dimension dominates.

⁷We note that our empirical approach in Equation 3 is different from taking the average of the individually estimated coefficient for each country, which has been documented to generally yield inconsistent estimates (see, e.g., [Pesaran and Smith \(1995\)](#)).

⁸As noted by [Wooldridge \(2011\)](#).

⁹While not central to our analysis, we also document, in the Appendix, that our estimates using the baseline dynamic panel regression are consistent with the GMM-based [Arellano and Bond \(1991\)](#) estimator with lags of up to 12 months included in the instrument set.

Our choice of controls in Equation 3 builds heavily on the literature. As noted by [Schwert \(1990\)](#), uncertainty arising from the stock market rose, on average, during recessions using pre-1987 data for the United States. More recently, [Reinhart and Rogoff \(2008\)](#) document that financial crises are typically followed by a large and persistent decline in economic performance. Along this dimension, we control for uncertainty arising from the stock market (via the VIX index), from the commodity market (via oil volatility), from changes in policy (via the [Baker et al. \(2016\)](#) EPU index), and from falling aggregate demand (via the consumer confidence index - CCI from the OECD).

We include two lags to account for delays in the impacts of these variables. Our choice of two lags is based on the most common lag length when tested (according to the SBIC) using individual country regressions on the full set of controls and a max lag length of 12 months. Such an optimal lag length of two months is consistent whether we use a max lag length of 24 or 8 months. We also test for unit roots in selected variables using the [Im et al. \(2003\)](#) panel unit roots test and provide the corresponding statistics in the Appendix. Overall, we reject the null hypothesis that all cross-sections contain unit-roots.

3.2 Results

Stock Market Dispersion and the Unemployment Rate We present the regression results using Equation 3, with the unemployment rate as the dependent variable, in Table 2. From left to right, we start with the regression specification with stock market dispersion on the unemployment rate and then gradually add the changes in the interest rate (Specification (2)), the changes in MCI (Specification (3)), the changes in CCI (Specification (4)), the changes in EPU (Specifications (5)-(7)), the VIX index (Specification (8)), and oil volatility (Specification (9)).

[Table 2 Here]

Turning to more details, the coefficients of stock market dispersion on unemployment are positive and significant (for the first lag) across all specifications. We interpret this as evidence that increases in stock market dispersion are significantly associated with negative movements

in the labor market. In principle, stock market dispersion might capture changes in both labor market structures and the *levels* of aggregate demand, per [Abraham and Katz \(1986\)](#)'s critique. Since changes in aggregate demand are reflected in the lags of the unemployment rate and CCI, we find it reassuring that our results are consistent even after controlling for these lags. Our results in Table 2 are also robust, in terms of statistical significance and the signs of the coefficients, when we drop the U.S. from the sample.

Intuitively, since stock market returns are indicative of investors' expectation of a firm's future profits, an increase in returns dispersion is indicative of a shift from one industry to another. When investors re-balance their portfolios from the declining industries to the expanding ones, we should expect an increase in the cross-industry standard deviation of stock market returns, which is reflective of shifts in sector demand. As aptly pointed out by [Lilien \(1982\)](#) and [Davis \(1987\)](#), these shifts imply significant costs to both the firms and the workers, which result in higher unemployment in the short run. Empirically, our result dovetails with [Loungani et al. \(1990\)](#), who document a significant and negative relationship between lags of stock market dispersion across U.S. industries and employment.

Stock Dispersion and Output Growth Table 3 presents the results estimated using Equation 3, but with IP growth instead of the unemployment rate as the dependent variable. Similar to Table 2, here we start with a basic specification involving stock market dispersion on IP growth and then gradually add our set of controls. The key result here is that the coefficients for the lags of stock market dispersion on IP growth are both negative and statistically significant across the first and second lags. This is consistent with the view that an increase in stock market dispersion across industries is generally associated with depressed economic activity in the short run. Our result is not surprising, given the co-movement of unemployment and output growth and the significant relationship between stock market dispersion and unemployment documented in Table 2.

[Table 3 Here]

A natural question arising from our result is that, since we construct our measure of dispersion from stock market returns, how much information from this measure is simply reflective of stock market uncertainty. As carefully documented in the literature (most prominently [Schwert \(1990\)](#)), periods of high stock market volatility typically precede periods of low economic performance. Along this dimension, we find that the coefficients of stock market dispersion are significant and negative, controlling for uncertainty arising from the stock market (using the VIX index), from changes in economic policy (using the EPU index), and from consumer confidence - or the lack thereof (using the MCI and CCI). The consistency of the estimated coefficients across these specifications suggests that stock market dispersion is able to explain movements in IP growth beyond what these measures of uncertainty are able to.¹⁰ Another related issue is that since countries are different in terms of labor market frictions, it is possible that our estimated effects (on both unemployment and IP growth) are driven by such differences. We examine this possibility using the Labor Market Efficiency Index from the World Economic Forum and find that our results are generally consistent even after controlling for labor market frictions across countries.¹¹

3.3 Subsample Analyses

We next divide our sample into (i) a sub-sample that starts from the beginning of the Great Recession (2007M12-2017M5) and (ii) a sub-sample that starts from the beginning of the Dot-com bubble (2001M3-2017M5). Our goal here is to understand the extent to which the result that increases in stock market dispersion depress economic activity varies across different sample periods. To that end, we find the starts of the last two U.S. recessions to be natural dividers for our sub-sample analysis.

¹⁰Since the VIX index enters the equation linearly, it is possible that the index might not be able to fully capture uncertainty arising from the stock market in a non-linear fashion. We address this issue first taking the natural log of the VIX index, as suggested by an anonymous referee. In Tables C.3 and C.4 the appendix, we report that our results are consistent regardless of whether this uncertainty measure enters the equation linearly.

¹¹Data for the Labor Market Efficiency Index are from the World Economic Forum Global Competitiveness Index. Such data are of annual frequency and are available from 2007 to 2017. To economize on space, we leave the details of this exercise with the Labor Market Efficiency Index as a control in Table C.5 in the appendix.

[Table 4 Here]

We next present the results using specifications (4)-(9) from Tables 2 and 3, but with these two sub-samples in Tables 4 and 5. Our estimated coefficients for stock market dispersion on IP growth (and the unemployment rate) are both statistically significant and negative (positive for the table that uses the unemployment rate). This consistency between the estimates in Tables 4 and 5 and the estimates in Tables 2 and 3 suggests that our results are not sensitive to these two different sample periods.

[Table 5 Here]

Comparing the coefficients across these two subsamples yields a key insight. During the periods following the Great Recession (2007M12-2017M5), the coefficients of stock market dispersion on both the unemployment rate and IP growth are generally more pronounced (in magnitude) than the same coefficients using the sample that also covers the periods before the Great Recession (2001M3-2017M5). Given that stock market dispersion captures the re-allocation of resources from one industry to another (as in [Loungani et al. \(1990\)](#)), we find this result consistent with the literature that documents large structural changes in the labor market during the Great Recession (e.g., [Elsby et al. \(2010\)](#)).

4 Panel VAR Evidence

We next examine the extent to which the unemployment rate and IP growth respond to an unexpected increase in stock market dispersion using panel VARs that control for unobserved heterogeneity arising from country-specific characteristics and cross-country spillovers. We start with a panel VAR that mimics our dynamic panel regression (Specification (2) in Tables 2 and 3) and then proceed to the specifications that include the remaining controls.

Overall, we find unemployment to respond positively following an unexpected increase in stock market dispersion, while IP growth responds negatively to the same shock. We also find these responses to be persistent, lasting up to 20 months in some cases. These results dovetail

with the findings of [Loungani \(1986\)](#), [Davis \(1987\)](#), and more recently [Chen et al. \(2011\)](#) in the sense that increases in stock market dispersion, as a proxy for sector shifts, signal depressed economic activity.

4.1 Panel VAR Estimation

We estimate the following panel VAR system, following [Love and Zicchino \(2006\)](#), using GMM

$$Y_{it} = \alpha + \sum_{k=1}^2 \beta_k Y_{it-k} + f_i + d_t + \varepsilon_{it}, \quad (4)$$

in which Y_{it} includes (1) an indicator of economic performance (either the change in the unemployment rate or IP growth), (2) the change in the interest rate, and (3) stock market dispersion, in such an order, as follows

$$Y_{it} = [Unemp, \Delta Rate, Dispersion]' \quad \text{or} \quad Y_{it} = [IP, \Delta Rate, Dispersion]' \quad (5)$$

The country fixed effects (f_i) are included to control for country-specific heterogeneity. Since these fixed effects are usually correlated with the regressors, we use a forward mean-differencing procedure with up to 6 lagged regressors as instruments for our GMM estimation.¹² In addition to f_i , we also allow for time dummies d_t that are common across all countries. Including f_i and d_t allows us to control for unobserved heterogeneity arising from both country-specific characteristics and cross-country spillovers (i.e., international linkages), respectively. This empirical strategy is motivated by the recent empirical evidence that macroeconomic fundamentals across the world are typically driven by a common global dynamic factor (see, e.g., [Crucini et al. \(2011\)](#)).

Our identification strategy is to use a Cholesky decomposition with a lower triangular matrix restriction and the same ordering as in Equation 5. This setup implies variables that appear later in the ordering effect variables that appear earlier with a lag. Our identification strategy builds

¹²We also test the sensitivity of our results using sets of instruments up to 12 lags and find that our results are qualitatively consistent across these sets of instruments.

on the notion that financial markets typically respond almost immediately to any unexpected change in the economy, which the unemployment rate and IP growth are intended to capture. On the other hand, since firms need time to adjust to changes in the state of the economy, unexpected movements in the stock market are typically reflected in the unemployment rate and IP growth with a delay.¹³ For the rest of the section, we generate 95% confidence bands for the IRFs using 2,000 replications.

4.2 Impulse Responses to Stock Market Dispersion

Stock Market Dispersion and the Unemployment Rate We present the impulse responses of the unemployment rate and the change in the interest rate to a positive one-standard-deviation shock to stock market dispersion in the upper sub-figure of Figure 2. On top of each set of impulse responses, we indicate the ordering of the variables used in Y_{it} . Since we do control for country-specific and cross-country heterogeneity, these impulse responses can be interpreted as the average effects of stock market dispersion on the unemployment rate across the 16 countries in our sample.

[Figure 2 Here]

Overall, we find an increase in stock market dispersion to have a negative impact on the labor market in the short term, as evidenced by the positive response of the unemployment rate. Specifically, the unemployment rate increases by 0.02% following a positive one-standard-deviation shock to stock market dispersion. This increase peaks after the seventh month and gradually fade away after 15 to 20 months. Despite its modest magnitude at peak (0.02%), the impact of stock market dispersion on the unemployment rate is relatively long-lived, with the responses lasting beyond the 20-month horizon. Our result is consistent with previous studies that use U.S. data (e.g., [Loungani et al. \(1990\)](#), and more recently [Angelidis et al. \(2015\)](#)) in the sense that unemployment significantly depends on the lags of stock market dispersion.

¹³We find that our results are not sensitive to this ordering. We replicate, in Figure C.1 in the Appendix, these responses with an alternative ordering.

Stock Market Dispersion and IP Growth In the lower sub-figure of Figure 2, we plot the impulse responses of IP growth to a positive one-standard-deviation shock to stock market dispersion. Turning to more details, an unexpected increase in stock market dispersion has a negative impact on IP growth, and this impact is especially more pronounced after 5 months. IP growth decreases, at its lowest point, by roughly 0.4% following a positive one-standard-deviation shock to stock market dispersion. Compared to unemployment, IP growth appears to be more responsive (in terms of the peak magnitude) to an unexpected increase in stock market dispersion.

[Figure 3 Here]

Another insight is that the response of IP growth is relatively long-lived, with the negative impact lasting at least 15 months. Since increases in stock market dispersion can be thought of as signals of sector shifts (e.g., [Loungani et al. \(1990\)](#)), this persistent effect is rather unsurprising. Intuitively, as investors re-allocate resources from one industry to another - as measured by an increase in stock market dispersion - workers are typically faced with significant transition costs. These transition costs can in turn ramp up the downward pressure on aggregate demand, as predicted, for example, by the seminal works of [Lucas and Prescott \(1974\)](#) and [Hall \(1979\)](#). The overall impact of this chain of events is that an increase in stock market dispersion can signal a reduction in future economic performance.

Forecast Error Variance Decomposition We next report the Forecast Error Variance Decomposition (FEVD) for our panel VARs using the unemployment rate and IP growth in Figure 3. Specifically, here we plot the contribution of stock market dispersion and the change in the interest rate to variation in unemployment and IP growth up to 20 months following the initial shock. Excluding unemployment/IP growth's lags, we find that stock market dispersion is able to explain roughly a third (for the unemployment VAR, the upper sub-figure) or half (for the IP growth VAR, the lower sub-figure) of the variation in unemployment and IP growth, respectively. Moreover, the contribution of stock market dispersion to the variation in unemployment and IP

growth starts to accelerate in the 5th month and continue to rise well beyond the 15-month horizon.

Granger-Causality Test We next document the relevant statistics for the Granger causality tests in Table 6, with adjustments for panel data following [Love and Zicchino \(2006\)](#) and [Dumitrescu and Hurlin \(2012\)](#). Overall, we find causality running from stock market dispersion to unemployment and IP growth to be significant. Moreover, the adjusted test statistics suggest that the direction of stock market dispersion Granger-causing unemployment and IP growth is *much stronger* than the other way around. This result is rather unsurprising given the dynamic panel analysis, in which we find stock market dispersion to significantly impact future unemployment and IP growth. In the context of [Abraham and Katz \(1986\)](#)’s critique that measures of employment dispersion might reflect both sector shifts and movements in aggregate demand, it is reassuring to find stock market dispersion is more likely to imply movements in unemployment and IP growth than the other way around.

[Table 6 Here]

4.3 Alternative Specifications: Panel VARs

We next explore a number of alternative specifications to the baseline panel VAR in Section 4.1 by considering the rest of the control variables used in Section 3. Specifically, we re-estimate the following panel VAR

$$Y_{it} = \alpha + \sum_{k=1}^2 \beta_k Y_{it-k} + f_i + d_t + \varepsilon_{it}, \quad (6)$$

in which Y_{it} denotes a four-variable vector of the following: (1) a measure of economic performance (either IP growth or unemployment), (2) the change in the interest rate, (3) stock market dispersion, and (4) an additional control X , in such an order, as follows

$$Y_{it} = [Unemp, \Delta Rate, Dispersion, X]' \quad \text{or} \quad Y_{it} = [IP, \Delta Rate, Dispersion, X]' \quad (7)$$

Here X can be either (1) the CBOE:VIX (VIX) index, (2) oil volatility ($OilVolatility$), or (3) the change in the [Baker et al. \(2016\)](#) Economic Policy Uncertainty Index (ΔEPU). We next plot the impulse responses of unemployment/IP growth, the change in the interest rate, and X to a positive one-standard-deviation shock to stock market dispersion in Figures 4 and 5. To obtain these impulse responses and the corresponding confidence bands, we follow the same estimation and identification assumptions as in Section 4.1.

[Figure 4 Here]

Turning to more details, we find periods of high stock market dispersion to signal periods of depressed economic activity. This result is consistent when controlling for volatility arising from the stock market (via the VIX), from economic policy uncertainty (via the change in EPU), or from the commodity market (via oil volatility). In particular, IP growth appears to decrease (increase for unemployment) when there is a positive shock to stock market dispersion. This result is consistent with both our panel VAR and our dynamic panel regression results in the sense that economic performance is negatively dependent on the lags of stock market dispersion. The consistency between the three-variable panel VARs (in Section 4) and the four-variable panel VARs presented here dovetails with [Loungani et al. \(1990\)](#) and [Neumann and Topel \(1991\)](#), who find stock market dispersion to mostly capture structural changes.

[Figure 5 Here]

5 Conclusion

This paper examined the extent to which stock market dispersion is related to economic performance in a cross-country setting using dynamic panel regressions and panel VARs. Unlike earlier single-country-investigations, our empirical approach controls for unobserved heterogeneity (country-specific and from spillovers alike). Overall, we find significant evidence that economic activity (measured by unemployment and IP growth) responds negatively to stock market dispersion. Moreover, these results are consistent across various specifications, sub-samples, and

estimation methods. Our results confirm the sector shifts hypothesis that the unemployment rate is dependent on the lags of stock market dispersion that the previous literature has carefully documented in single-country settings (e.g., [Lilien \(1982\)](#) and [Loungani et al. \(1990\)](#)). Along this dimension, we provided cross-country evidence that such a hypothesis is empirically consistent across a dataset of 16 developed economies.

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Table 1: Descriptive Statistics (1994M1-2017M5)

Country	IP Growth		Dispersion		Unemployment (%)	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Austria	0.281	1.62	0.009	0.009	4.85	0.66
Belgium	0.242	2.45	0.008	0.004	8.20	0.93
Canada	0.131	0.99	0.008	0.003	7.60	1.13
Denmark	0.112	3.01	0.007	0.003	5.69	1.30
Finland	0.191	1.89	0.011	0.007	9.66	2.59
France	0.011	1.38	0.007	0.003	9.91	1.46
Germany	0.161	1.52	0.008	0.004	7.73	1.46
Ireland	0.831	6.01	0.011	0.007	9.01	4.06
Italy	-0.011	1.35	0.007	0.003	9.66	1.92
Japan	0.041	1.92	0.006	0.002	3.84	0.76
Netherlands	0.121	2.19	0.009	0.004	5.49	1.45
Norway	0.021	3.01	0.010	0.004	3.69	0.71
Spain	0.051	1.46	0.008	0.012	16.59	5.62
Sweden	0.161	2.04	0.009	0.004	7.54	1.23
U.K.	0.011	0.90	0.007	0.003	6.31	1.42
U.S.	0.151	0.65	0.006	0.003	5.91	1.62
Total	1.56	2.37	0.008	0.005	7.62	3.74

Note: This table presents the mean and the standard deviation of monthly stock market dispersion, IP growth, and the harmonized unemployment rate from the OECD from 1994M1 to 2017M5. Data are of monthly frequency and are seasonally adjusted.

Table 2: Dynamic Panel Regression: the Unemployment Rate

	Dependent Variable: the Unemployment Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	1.383*** (2.96)	1.061** (2.50)	1.056** (2.54)	1.077** (2.54)	1.078** (2.53)	1.033** (2.38)	1.025** (2.34)	1.104** (2.17)	1.149** (2.19)
Stock Market Dispersion (2-lag)	0.852** (2.18)	0.338 (1.13)	0.362 (1.20)	0.418 (1.31)	0.416 (1.29)	0.434 (1.32)	0.433 (1.31)	0.366 (1.21)	0.324 (1.02)
Unemployment (1-lag)	0.984*** (6.89)	0.960*** (6.63)	0.960*** (6.62)	0.960*** (6.34)	0.960*** (6.33)	0.960*** (6.34)	0.960*** (6.34)	0.960*** (6.32)	0.957*** (6.26)
Unemployment (2-lag)	-0.070 (-0.51)	-0.049 (-0.35)	-0.050 (-0.36)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.056 (-0.39)
Rate (1-lag)		-0.072*** (-6.93)	-0.073*** (-7.07)	-0.074*** (-6.71)	-0.075*** (-6.70)	-0.076*** (-6.94)	-0.076*** (-6.95)	-0.074*** (-6.46)	-0.073*** (-6.18)
Rate (2-lag)		-0.069** (-2.58)	-0.068** (-2.55)	-0.076** (-2.78)	-0.076** (-2.82)	-0.076** (-2.74)	-0.076** (-2.75)	-0.075** (-2.73)	-0.076** (-2.71)
MCI (1-lag)			-0.000 (-0.46)	-0.000 (-0.11)	-0.000 (-0.15)	-0.000 (-0.32)	-0.000 (-0.34)	-0.000 (-0.74)	-0.000 (-0.60)
MCI(2-lag)			0.001* (1.87)	0.001 (1.47)	0.001 (1.42)	0.001* (1.79)	0.001 (1.75)	0.001* (2.09)	0.001* (1.89)
CCI (1-lag)				0.011 (0.40)	0.010 (0.40)	0.010 (0.40)	0.010 (0.40)	0.007 (0.26)	0.006 (0.23)
CCI(2-lag)				-0.003 (-0.14)	-0.003 (-0.13)	-0.002 (-0.12)	-0.002 (-0.11)	0.000 (0.01)	-0.004 (-0.19)
EPU - EU (1-lag)					0.000 (0.07)	0.000 (0.08)	0.000 (0.19)	0.000 (0.26)	0.000 (0.06)
EPU - EU (2-lag)					0.000 (0.25)	0.000 (0.44)	0.000 (0.42)	0.000 (0.33)	0.000 (0.02)
EPU - US (1-lag)						0.000 (0.59)	0.000 (0.58)	0.000 (0.46)	0.000 (0.40)
EPU - US (2-lag)						-0.000 (-1.68)	-0.000 (-1.75)	-0.000* (-1.79)	-0.000* (-1.99)
EPU - Japan (1-lag)							-0.000 (-1.04)	-0.000 (-1.35)	-0.000 (-1.35)
EPU - Japan (2-lag)							-0.000 (-0.32)	-0.000 (-0.32)	-0.000 (-0.32)
VIX (1-lag)								-0.001 (-1.21)	-0.001 (-1.33)
VIX (2-lag)								0.000 (0.48)	0.000 (0.69)
Oil Volatility (1-lag)								-0.040* (-1.86)	-0.040* (-1.86)
Oil Volatility (2-lag)								-0.027 (-1.19)	-0.027 (-1.19)
Constant	-0.001*** (-5.16)	-0.004*** (-5.59)	-0.004*** (-5.58)	-0.004*** (-5.41)	-0.004*** (-5.49)	-0.004*** (-5.38)	-0.004*** (-5.39)	-0.004*** (-5.35)	-0.003*** (-4.78)
Observations	4,368	4,269	4,269	3,990	3,990	3,990	3,990	3,990	3,990
Adj - R ²	0.148	0.152	0.152	0.154	0.154	0.154	0.154	0.154	0.154

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on the unemployment rate. The unemployment rate is the harmonized unemployment rate from the OECD, first-differenced. The interest rate (Rate), MCI, CCI, and all the EPUs are first-differenced. Stock market dispersion, the VIX index, and oil volatility are not first-differenced. Sample periods: 1994M1-2017M5. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table 3: Dynamic Panel Regression: IP Growth

	Dependent Variable: Industrial Production Growth								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	-0.035** (-2.51)	-0.032** (-2.31)	-0.031** (-2.28)	-0.028* (-2.06)	-0.028* (-2.07)	-0.029* (-2.08)	-0.028* (-2.06)	-0.027* (-1.90)	-0.026* (-1.80)
Stock Market Dispersion (2-lag)	-0.023*** (-5.46)	-0.017*** (-5.13)	-0.017*** (-5.05)	-0.015*** (-4.56)	-0.015*** (-4.54)	-0.014*** (-4.71)	-0.014*** (-4.72)	-0.012*** (-3.59)	-0.013*** (-3.52)
IP Growth (1-lag)	-0.326*** (-7.06)	-0.340*** (-8.08)	-0.341*** (-8.16)	-0.341*** (-7.57)	-0.341*** (-7.59)	-0.341*** (-7.60)	-0.341*** (-7.68)	-0.341*** (-7.75)	-0.345*** (-8.29)
IP Growth (2-lag)	-0.133*** (-3.52)	-0.148*** (-4.65)	-0.149*** (-4.66)	-0.145*** (-4.30)	-0.145*** (-4.33)	-0.145*** (-4.31)	-0.145*** (-4.37)	-0.146*** (-4.30)	-0.151*** (-5.00)
Rate (1-lag)		1.148*** (5.41)	1.136*** (5.38)	1.140*** (6.07)	1.164*** (5.90)	1.157*** (5.69)	1.167*** (5.81)	1.123*** (5.60)	1.155*** (5.70)
Rate (2-lag)		0.352* (1.90)	0.377* (2.10)	0.407** (2.15)	0.369* (2.02)	0.365* (1.88)	0.364* (1.88)	0.440** (2.34)	0.446** (2.30)
MCI (1-lag)			0.009*	0.000	0.002	0.002	0.002	-0.002	0.002
MCI(2-lag)			(1.92)	(0.10)	(0.51)	(0.33)	(0.47)	(-0.40)	(0.59)
CCI (1-lag)			0.015*	-0.003	-0.003	-0.002	-0.001	-0.011	-0.013*
CCI (2-lag)			(1.81)	(-0.36)	(-0.42)	(-0.27)	(-0.19)	(-1.54)	(-1.80)
				0.469	0.479	0.478	0.482*	0.448*	0.378
				(1.75)	(1.76)	(1.76)	(1.78)	(1.78)	(1.54)
				0.610***	0.601**	0.603**	0.594***	0.568***	0.429*
				(3.01)	(2.97)	(2.98)	(2.99)	(2.99)	(2.06)
EPU - EU (1-lag)				-0.000	-0.000	-0.001	-0.001	-0.001	-0.001
EPU - EU (2-lag)				(-1.05)	(-1.07)	(-1.07)	(-1.45)	(-1.71)	(-1.65)
EPU - US (1-lag)				-0.001*	-0.001*	-0.001*	-0.001*	-0.001*	-0.001**
EPU - US (2-lag)				(-1.88)	(-1.80)	(-1.80)	(-1.80)	(-1.76)	(-2.18)
EPU - Japan (1-lag)					0.000	0.000	0.000	0.001	0.001
EPU - Japan (2-lag)					(0.46)	(0.46)	(0.48)	(1.08)	(0.98)
VIX (1-lag)					-0.000	-0.000	-0.000	-0.000	-0.000
VIX(2-lag)					(-0.53)	(-0.53)	(-0.42)	(-0.23)	(-0.12)
Oil Volatility (1-lag)					0.001*	0.001*	0.001*	0.001*	0.001*
Oil Volatility (2-lag)					(2.10)	(2.10)	(2.10)	(1.95)	(1.86)
					(1.17)	(1.17)	(1.17)	(0.90)	(0.62)
								0.006	0.008
								(0.36)	(0.56)
								-0.036***	-0.038***
								(-3.25)	(-3.44)
									-0.452**
									(-2.29)
									-0.968***
									(-4.04)
Constant	0.228*** (16.92)	0.268*** (19.61)	0.268*** (19.73)	0.270*** (17.50)	0.269*** (17.32)	0.269*** (17.30)	0.270*** (17.24)	0.271*** (17.84)	0.282*** (21.07)
Observations	4,368	4,269	4,269	3,990	3,990	3,990	3,990	3,990	3,990
Adj - R ²	0.105	0.117	0.117	0.121	0.121	0.121	0.121	0.124	0.128

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on IP growth. The interest rate (Rate), MCI, CCI, and all the EPUs are first-differenced. Stock market dispersion, the VIX index, and oil volatility are not first-differenced. Sample periods: 1994M1-2017M5. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table 4: Dynamic Panel Regression with the Unemployment Rate: Different Sample Periods

	Dependent Variable: the Unemployment Rate					
	Sample: 2001M3-2017M5			Sample: 2007M12-2017M5		
Stock Market Dispersion (1-lag)	1.297** (2.12)	1.295** (2.13)	1.235** (2.00)	1.219** (1.95)	1.259** (1.74)	1.298* (1.73)
Stock Market Dispersion (2-lag)	0.453 (1.05)	0.455 (1.05)	0.451 (1.06)	0.443 (1.03)	0.389 (0.94)	0.341 (0.80)
Unemployment (1-lag)	0.950*** (5.68)	0.950*** (5.67)	0.951*** (5.69)	0.950*** (5.69)	0.950*** (5.68)	0.948*** (5.63)
Unemployment (2-lag)	-0.038 (-0.24)	-0.038 (-0.24)	-0.039 (-0.24)	-0.038 (-0.24)	-0.038 (-0.24)	-0.043 (-0.27)
Rate (1-lag)	-0.089*** (-3.21)	-0.088*** (-3.07)	-0.089*** (-3.11)	-0.089*** (-3.09)	-0.086*** (-3.01)	-0.086*** (-2.98)
Rate (2-lag)	-0.124** (-2.80)	-0.125** (-2.76)	-0.127** (-2.70)	-0.128** (-2.71)	-0.129** (-2.80)	-0.127** (-2.71)
MCI (1-lag)	0.000 (0.11)	0.000 (0.11)	-0.000 (-0.16)	-0.000 (-0.25)	-0.000 (-0.47)	-0.000 (-0.35)
MCI(2-lag)	0.001 (1.69)	0.001 (1.59)	0.001* (1.85)	0.001* (1.82)	0.001** (2.24)	0.001* (2.01)
CCI (1-lag)	-0.008 (-0.23)	-0.008 (-0.23)	-0.008 (-0.24)	-0.008 (-0.25)	-0.010 (-0.29)	-0.011 (-0.32)
CCI(2-lag)	0.013 (0.60)	0.013 (0.61)	0.014 (0.64)	0.014 (0.65)	0.016 (0.71)	0.011 (0.46)
EPU - EU (1-lag)		-0.000 (-0.31)	-0.000 (-0.34)	-0.000 (-0.17)	-0.000 (-0.07)	-0.000 (-0.29)
EPU - EU (2-lag)		0.000 (0.04)	0.000 (0.26)	0.000 (0.20)	0.000 (0.12)	-0.000 (-0.14)
EPU - US (1-lag)			0.000 (0.00)	0.000 (0.00)	0.000 (0.00)	0.000 (0.00)
EPU - US (2-lag)			0.91 (0.91)	0.92 (0.92)	0.76 (0.76)	0.73 (0.73)
EPU - Japan (1-lag)			-1.42 (-1.42)	-1.53 (-1.53)	-1.57 (-1.57)	-1.76 (-1.76)
EPU - Japan (2-lag)				-1.52 (-1.52)	-1.95 (-1.95)	-2.34 (-2.34)
VIX (1-lag)				-0.000 (-0.22)	-0.000 (-0.20)	-0.000 (-0.24)
VIX (2-lag)					-0.001 (-0.59)	-0.001 (-0.74)
Oil Volatility (1-lag)					0.000 (0.61)	0.001 (0.72)
Oil Volatility (2-lag)					-0.032 (-1.30)	-0.032 (-1.30)
Constant	-0.019** (-2.40)	-0.019** (-2.40)	-0.018** (-2.38)	-0.018** (-2.31)	-0.018* (-2.13)	-0.018* (-2.04)
Observations	2,909	2,909	2,909	2,909	2,909	2,909
Adj - R ²	0.164	0.164	0.164	0.164	0.164	0.164

Note: This table presents the results of stock market dispersion on the unemployment rate using dynamic panel regression for two different sample periods. The interest rate (Rate), MCI, CCI, and all the EPU are first-differenced. Stock market dispersion, the VIX index, and oil volatility are not first-differenced. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table 5: Dynamic Panel Regression with IP Growth: Different Sample Periods

	Dependent Variable: IP Growth											
	Sample: 2001M3-2017M5						Sample: 2007M12-2017M5					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
Stock Market Dispersion (1-lag)	-0.030 (-1.75)	-0.030 (-1.76)	-0.030* (-1.77)	-0.030 (-1.75)	-0.027 (-1.52)	-0.025 (-1.46)	-0.027 (-1.41)	-0.027 (-1.40)	-0.029 (-1.47)	-0.027 (-1.44)	-0.024 (-1.29)	-0.021 (-1.23)
Stock Market Dispersion (2-lag)	-0.012** (-2.68)	-0.012** (-2.68)	-0.012** (-2.72)	-0.012** (-2.66)	-0.009* (-1.97)	-0.011* (-2.09)	-0.023** (-2.30)	-0.023** (-2.23)	-0.022** (-2.21)	-0.022** (-2.17)	-0.019* (-2.12)	-0.021** (-2.39)
IP Growth (1-lag)	-0.352*** (-6.57)	-0.352*** (-6.61)	-0.351*** (-6.63)	-0.353*** (-6.67)	-0.353*** (-6.80)	-0.359*** (-7.48)	-0.248*** (-4.86)	-0.247*** (-4.88)	-0.246*** (-4.83)	-0.246*** (-4.94)	-0.245*** (-4.96)	-0.249*** (-5.19)
IP Growth (2-lag)	-0.158*** (-3.63)	-0.158*** (-3.65)	-0.158*** (-3.65)	-0.159*** (-3.71)	-0.159*** (-3.58)	-0.166*** (-4.29)	-0.152** (-2.28)	-0.150** (-2.29)	-0.151** (-2.30)	-0.155** (-2.39)	-0.153** (-2.33)	-0.163** (-2.76)
Rate (1-lag)	2.032*** (4.10)	2.057*** (4.08)	2.064*** (4.11)	2.067*** (4.15)	1.905*** (4.08)	1.927*** (4.29)	2.072*** (3.52)	2.104*** (3.54)	2.082*** (3.46)	2.093*** (3.51)	1.903*** (3.40)	1.951*** (3.56)
Rate (2-lag)	0.172 (0.55)	0.117 (0.38)	0.093 (0.29)	0.117 (0.36)	0.414 (1.39)	0.444 (1.46)	0.300 (0.99)	0.169 (0.52)	0.127 (0.35)	0.127 (0.35)	0.481 (1.65)	0.431 (1.45)
MCI (1-lag)	-0.002 (-0.29)	0.000 (0.07)	-0.001 (-0.11)	0.001 (0.23)	-0.006 (-0.59)	-0.001 (-0.09)	0.011 (0.85)	0.013 (1.03)	0.011 (0.85)	0.017 (1.13)	0.009 (0.45)	0.015 (0.77)
MCI(2-lag)	-0.006 (-0.76)	-0.006 (-0.77)	-0.005 (-0.66)	-0.004 (-0.48)	-0.014* (-1.83)	-0.017** (-2.16)	0.004 (0.33)	0.001 (0.06)	0.005 (0.29)	0.008 (0.45)	-0.000 (-0.03)	-0.008 (-0.48)
CCI (1-lag)	0.680** (2.61)	0.694** (2.66)	0.688** (2.64)	0.687** (2.65)	0.617** (2.71)	0.519** (2.28)	0.768** (1.83)	0.801* (1.94)	0.787* (1.96)	0.729* (1.76)	0.629* (1.85)	0.541 (1.56)
CCI(2-lag)	0.463* (2.10)	0.453* (2.12)	0.458* (2.12)	0.450** (2.18)	0.426** (2.22)	0.231 (0.99)	0.518 (1.52)	0.488 (1.51)	0.498 (1.55)	0.527 (1.67)	0.510* (1.79)	0.279 (0.82)
EPU - EU (1-lag)	-0.000 (-0.80)	-0.000 (-0.80)	-0.000 (-0.89)	-0.001 (-1.28)	-0.001 (-1.45)	-0.001 (-1.44)	-0.001 (-1.51)	-0.001 (-1.51)	-0.001 (-1.22)	-0.002** (-2.41)	-0.001** (-2.50)	-0.001** (-2.36)
EPU - EU (2-lag)	-0.001 (-1.74)	-0.001 (-1.74)	-0.001* (-1.86)	-0.001* (-1.97)	-0.001* (-1.89)	-0.001** (-2.52)	-0.001* (-2.01)	-0.001* (-2.01)	-0.001** (-2.18)	-0.001** (-2.19)	-0.001* (-2.03)	-0.001** (-2.43)
EPU - US (1-lag)	0.000 (0.60)	0.000 (0.60)	0.000 (0.60)	0.000 (0.59)	0.001 (0.98)	0.000 (0.88)	0.000 (0.98)	0.000 (0.98)	0.000 (0.74)	0.000 (0.90)	0.001 (1.42)	0.001 (1.39)
EPU - US (2-lag)	-0.000 (-0.23)	-0.000 (-0.23)	-0.000 (-0.23)	-0.000 (-0.11)	-0.000 (-0.11)	0.000 (0.11)	-0.001 (-0.75)	-0.001 (-0.63)	-0.001 (-0.75)	-0.001 (-0.63)	-0.001 (-0.71)	-0.000 (-0.42)
EPU - Japan (1-lag)	0.001 (1.76)	0.001 (1.76)	0.001 (1.76)	0.001 (1.76)	0.001 (1.39)	0.001 (1.38)	0.001 (1.38)	0.001 (1.38)	0.001 (1.38)	0.001** (2.79)	0.001* (2.02)	0.001 (1.60)
EPU - Japan (2-lag)	0.001** (2.60)	0.001** (2.60)	0.001** (2.60)	0.001** (2.60)	0.000 (1.63)	0.000 (1.36)	0.000 (1.63)	0.000 (1.63)	0.000 (1.36)	0.001** (2.73)	0.001* (2.06)	0.001 (1.60)
VIX (1-lag)	-0.006 (-0.26)	-0.006 (-0.26)	-0.006 (-0.26)	-0.006 (-0.26)	-0.006 (-0.26)	-0.003 (-0.14)	-0.006 (-0.26)	-0.006 (-0.26)	-0.006 (-0.26)	-0.008 (-0.31)	-0.008 (-0.31)	-0.001 (-0.05)
VIX (2-lag)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.041*** (-3.63)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.037*** (-3.43)	-0.034** (-2.82)	-0.042** (-2.82)
Oil Volatility (1-lag)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)	-0.524** (-2.27)
Oil Volatility (2-lag)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)	-1.051*** (-4.49)
Constant	0.179*** (12.94)	0.177*** (12.74)	0.177*** (12.75)	0.177*** (12.78)	0.177*** (13.22)	0.182*** (14.11)	0.118*** (5.60)	0.113*** (5.42)	0.110*** (5.23)	0.111*** (5.39)	0.115*** (6.14)	0.113*** (6.02)
Observations	2,849	2,849	2,849	2,849	2,849	2,849	1,650	1,650	1,650	1,650	1,650	1,650
Adjusted - R ²	0.132	0.133	0.132	0.133	0.135	0.141	0.097	0.098	0.099	0.103	0.105	0.112

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on IP growth for two different sample periods. The interest rate (Rate), MCI, CCI, and all the EPUs are first-differenced. Stock market dispersion, the VIX index, and oil volatility are not first-differenced. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table 6: Panel VAR-Granger causality Wald test

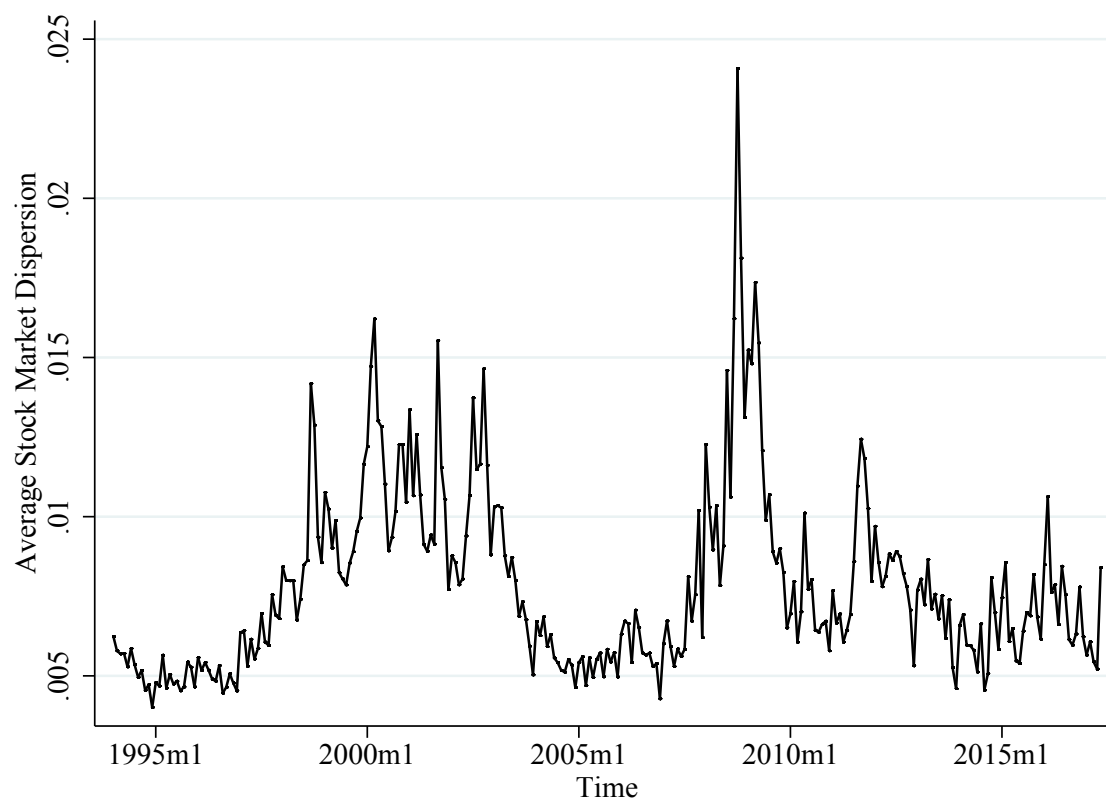
Dispersion $\rightarrow$ Unemployment			Dispersion $\rightarrow$ IP Growth		
Lags	χ^2	p	Lags	χ^2	p
1	6.560	0.010	1	9.034	0.003
2	10.156	0.006	2	16.639	0.000

Unemployment $\rightarrow$ Dispersion			IP Growth $\rightarrow$ Dispersion		
Lags	χ^2	p	Lags	χ^2	p
1	0.325	0.568	1	5.660	0.017
2	3.303	0.192	2	6.726	0.035

Ho: Variable on the left does not Granger-cause the variable on the right

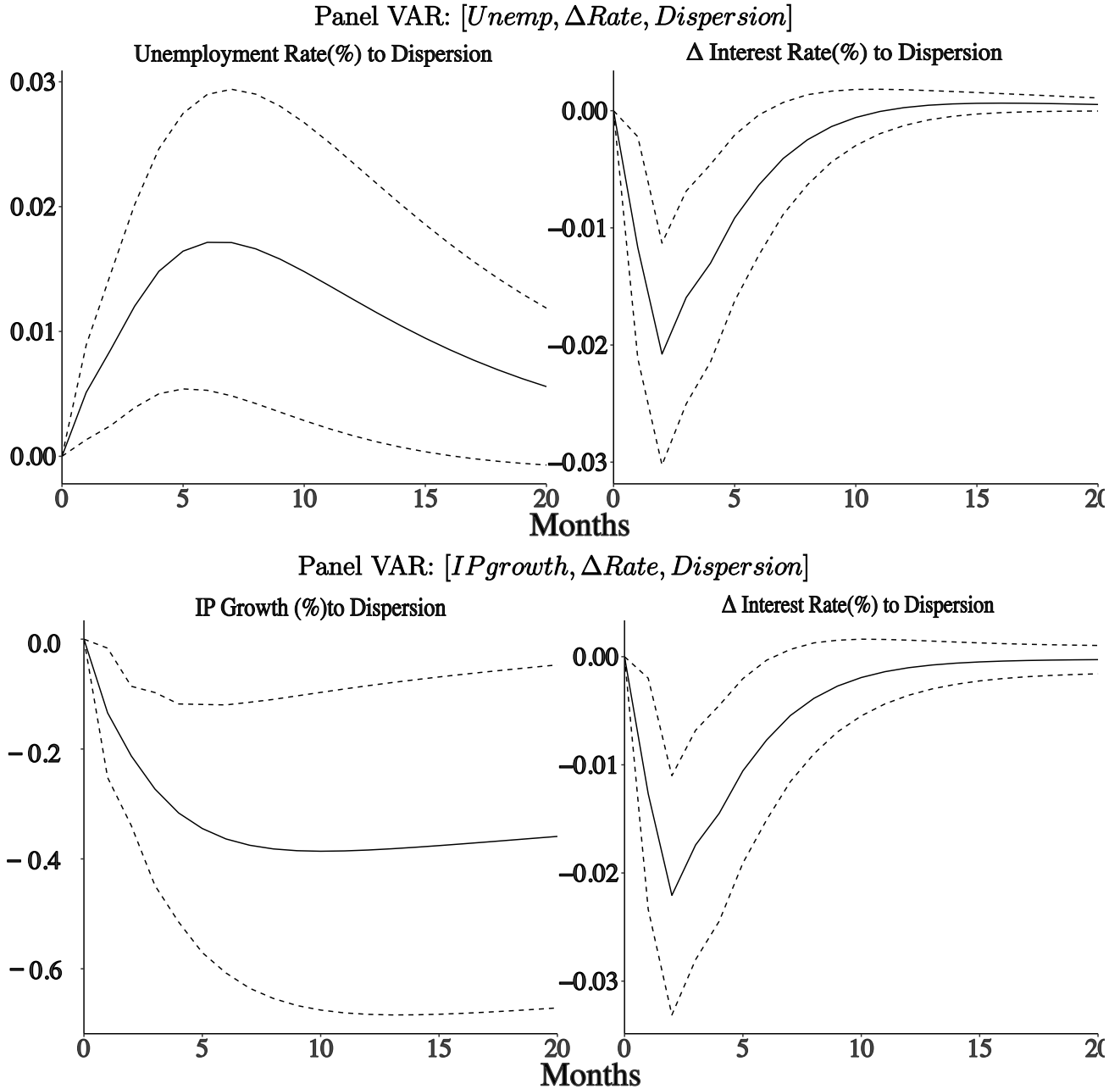
Ha: Variable on the left Granger-causes the variable on the right

Figure 1: Average Stock Market Dispersion across Countries



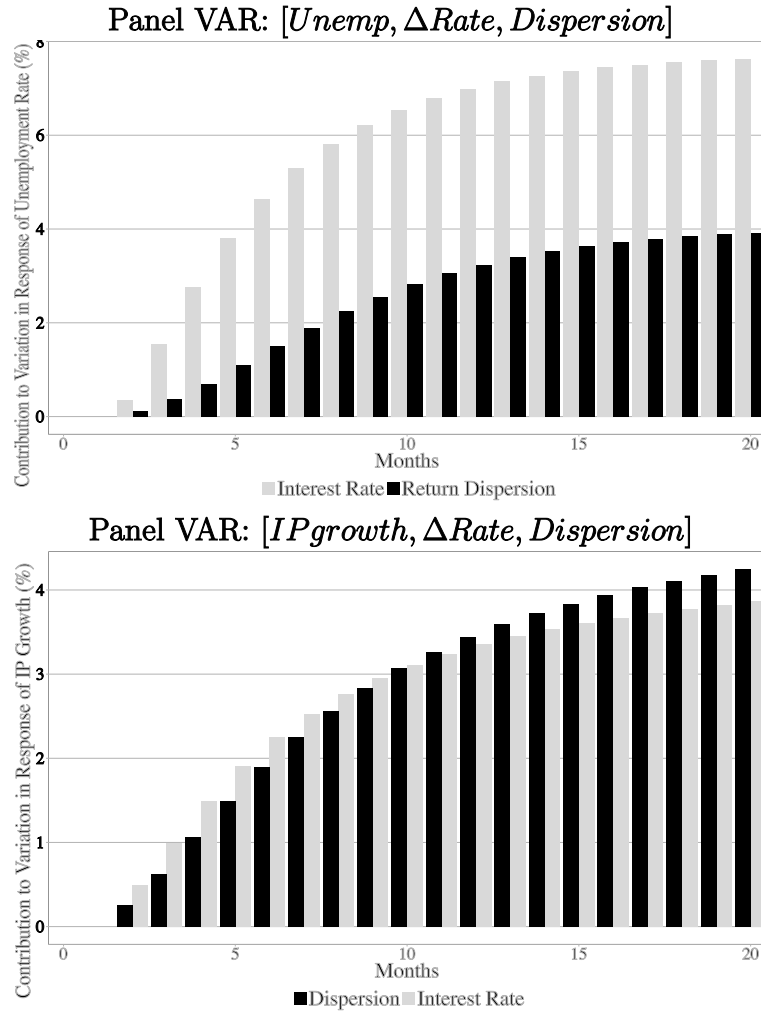
Note: This figure shows the average stock market dispersion across countries from 1994M1 to 2017M5.

Figure 2: Unemployment, IP Growth and Interest Rate to Dispersion Shock



Note: This figure presents the impulse responses of IP growth (in percentage) and the interest rate (in percentage) to a positive one-standard-deviation shock to stock market dispersion. Confidence bands (95%) are generated using 2,000 replications.

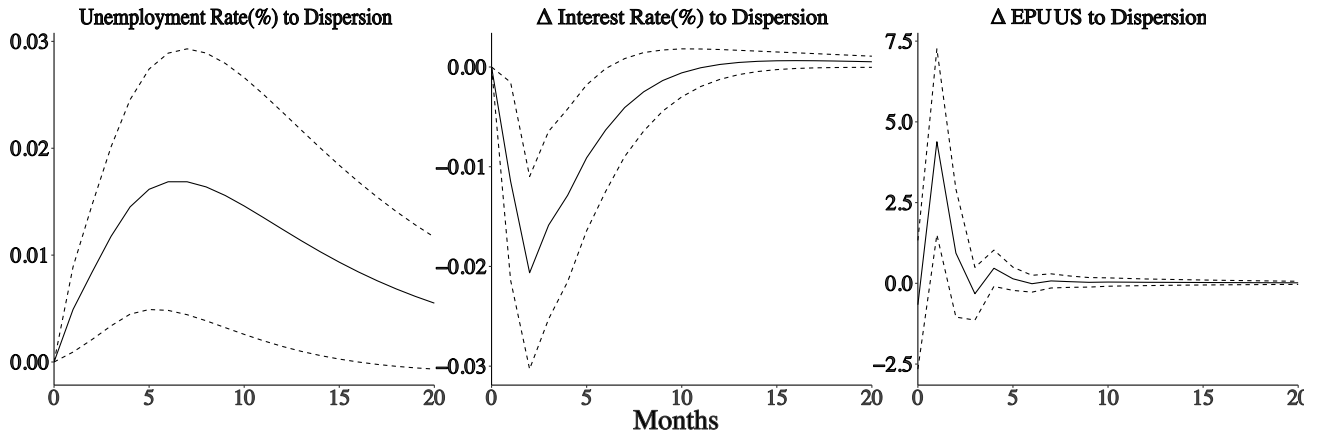
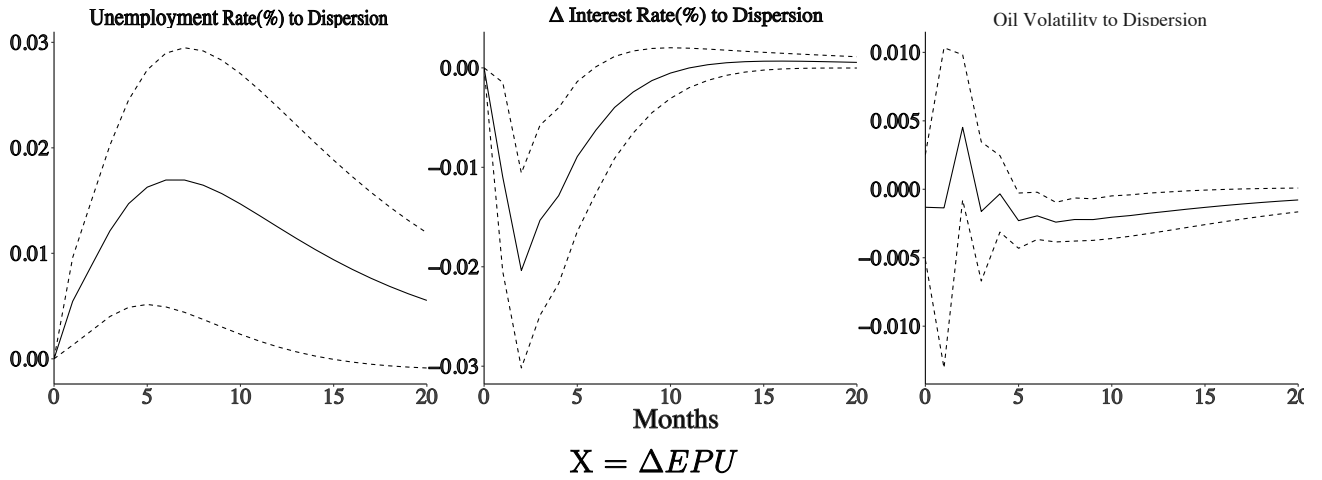
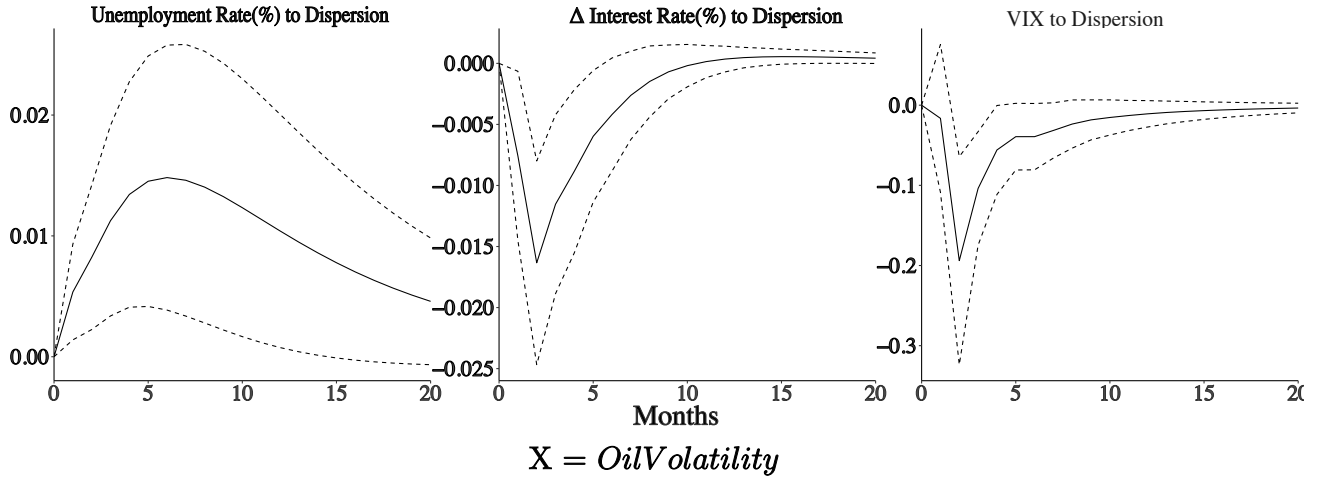
Figure 3: Forecast Error Variance Decomposition



Note: These plots present the percentage contribution of the change in the interest rate and stock market dispersion to variation in the unemployment rate (top) and IP growth (bottom) using the panel VAR with the unemployment rate $Y_{it} = [Unemp, \Delta Rate, Dispersion]$ (top) and the panel VAR with IP growth $Y_{it} = [IPgrowth, \Delta Rate, Dispersion]$ (bottom).

Figure 4: Responses to Dispersion Shock: the Unemployment Rate

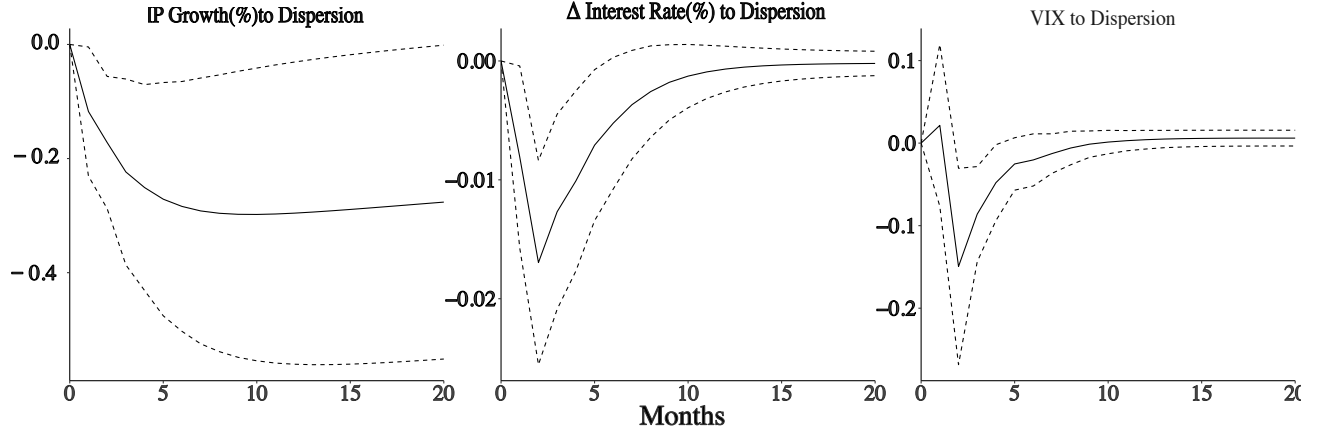
Panel VAR Order: $[Unemp, Dispersion, \Delta Rate, X]$
 $X = VIX$



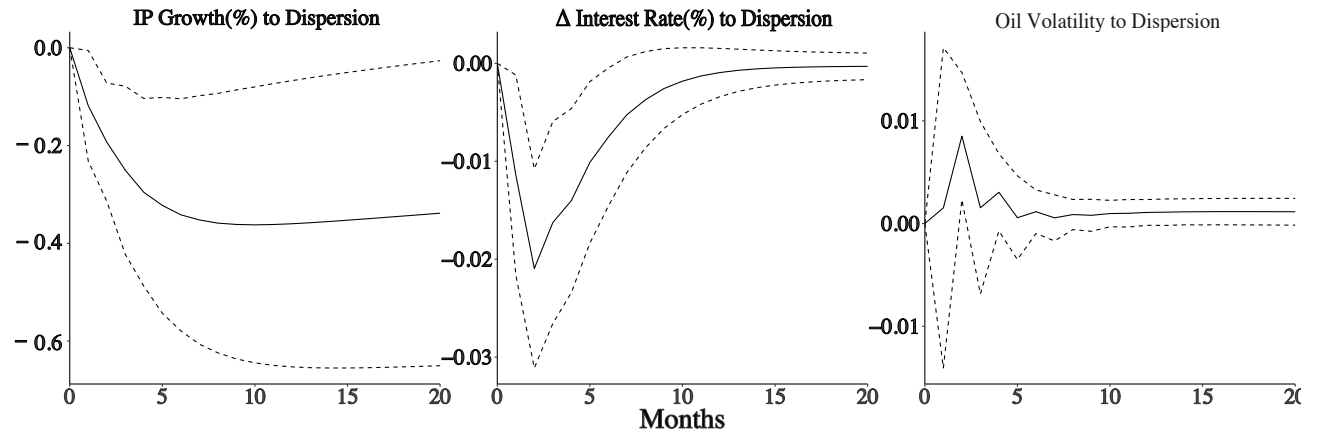
Note: This figure presents the impulse responses of the unemployment rate (in percentage), the interest rate (in percentage), and X to a positive one-standard-deviation shock to stock market dispersion. Here X is either (1) VIX (top), (2) oil volatility (middle), or (3) the change in the [Baker et al. \(2016\)](#) index for the U.S. (bottom). Confidence bands (95%) are generated using 2,000 replications.

Figure 5: Responses to Dispersion Shock: IP Growth

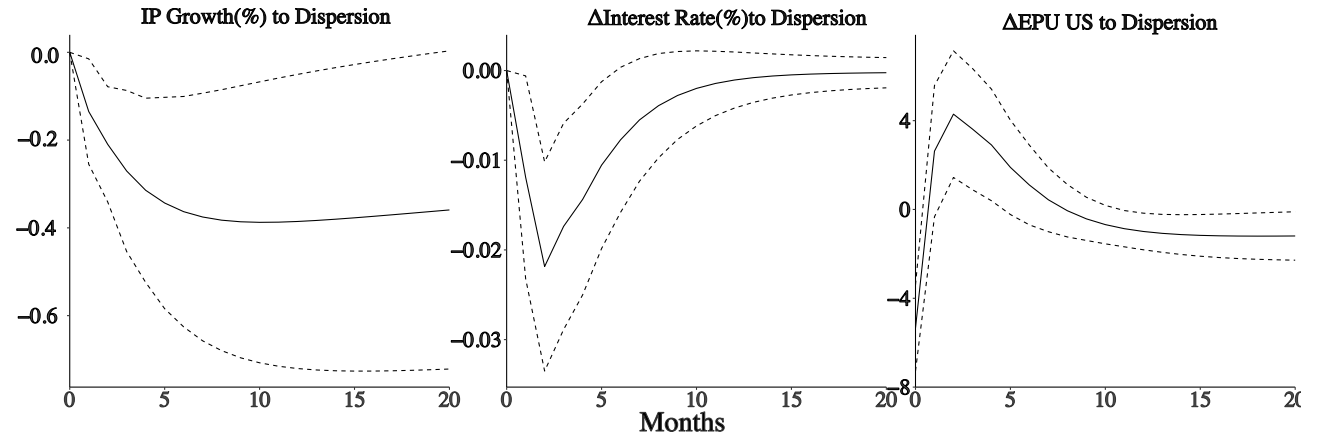
Panel VAR Order: $[IPGrowth, Dispersion, \Delta Rate, X]$
 $X = VIX$



$X = OilVolatility$



$X = \Delta EPU$



Note: This figure presents the impulse responses of IP growth (in percentage), the interest rate (in percentage), and X to a positive one-standard-deviation shock to stock market dispersion. Here X is either (1) VIX (top), (2) oil volatility (middle), or (3) the change in the [Baker et al. \(2016\)](#) index for the U.S. (bottom). Confidence bands (95%) are generated using 2,000 replications.

Appendix: Not for Publication

A Data

The FTSE 350 Index is used to construct our measure of stock market dispersion and is from Datastream. The index is a market-capitalization-weighted stock market index that includes the largest 350 companies by capitalization which have their primary listing on the London Stock Exchange. The FTSE 350 is a combination of the FTSE 100 Index of the largest 100 companies and the FTSE 250 Index of the next largest 250, representing large- and mid-cap stocks traded on the London Stock Exchange, which pass screening for size and liquidity. The index is maintained by FTSE Russell, a subsidiary of the London Stock Exchange Group.¹⁴

$$\text{Index level} = \frac{\sum_i \text{Price of stock}_i \times \text{Number of shares}_i \times \text{Free float adjustment factor}_i}{\text{Index divisor}} \quad (8)$$

The Free float Adjustment factor measures the percentage of all issued shares that are readily available for trading. The factor is then rounded up to the nearest multiple of 5 percents. The twelve sectors are chosen for the calculation of stock market dispersion, including banking, electricity, finance, mass media, manufacturing, consumer goods, mining, technology, telecommunications, health care, oil and gas, and utilities.

¹⁴These descriptions are from Thompson Reuters. Last accessed July 01, 2018.

Table A.1: A Summary of Variables and Sources

Variable	Source	Time Period
FTSE Stock Indexes	Thompson Reuters	Jan. 1994 - May. 2017
Industrial Production Growth	OECD	Jan. 1994 - May. 2017
Short-term (3-month) Interest Rate	OECD	Jan. 1994 - May. 2017
Economic Policy Uncertainty (EPU) Index	Baker et al. (2016)	Jan. 1994 - May. 2017
West Texas Intermediate (WTI) Oil Price	St. Louis' FRED	Jan. 1994 - May. 2017
Michigan Consumer's Sentiment Index (MCI)	University of Michigan's Institute for Social Research	Jan. 1994 - May. 2017
Consumer's Confidence Index (CCI)	OECD	Jan. 1994 - May. 2017
VIX Index	CBOE	Jan. 1994 - May. 2017
Harmonized Unemployment Rate	St. Louis' FRED	Jan. 1994 - May. 2017

B Panel Unit Root Test

[Im et al. \(2003\)](#) develop a set of tests that relax the assumption of a common autoregressive parameter. Under the null hypothesis, all panels contain unit roots. Based on the test statistics from Table B.1, we strongly reject the null hypothesis that all panels contain unit roots.

Table B.1: [Im et al. \(2003\)](#) Tests for Stationarity for Main Variables

Variable	Z-Statistic	p-Value
IP Growth	-59.64	0.000
Stock Market Dispersion	-39.56	0.000
Unemployment Rate	-58.38	0.000
Ho: All panels contain unit roots.		
Ha: Some panels are stationary.		

C Estimation using the [Arellano and Bond \(1991\)](#) Estimator

In this section, we re-estimate the baseline dynamic panel regression

$$Y_{it} = \alpha + \sum_{k=1}^K \beta_k Y_{it-k} + \sum_{k=1}^K \gamma_k X_{it-k} + \varepsilon_{it} \quad (9)$$

using the [Arellano and Bond \(1991\)](#) estimator with up to 6 lags as instruments. While not reported, we also estimate Equation 9 with 6-12 lags as instruments and find that our results are qualitatively and quantitatively robust across specifications using these additional numbers of lags. As shown in Tables C.1 and C.2, our estimates for the impacts of stock market dispersion on IP growth and unemployment are both statistically significant and consistent with the narrative that increases in stock market dispersion depress economic activity.

Table C.1: Dynamic Regression Results using IP Growth: With the [Arellano and Bond \(1991\)](#) Estimator

	Dependent Variable: IP Growth								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	-0.029*** (-4.02)	-0.028*** (-3.83)	-0.027*** (-3.75)	-0.020*** (-2.83)	-0.020*** (-2.84)	-0.021*** (-2.88)	-0.020*** (-2.80)	-0.017** (-2.39)	-0.016** (-2.23)
Stock Market Dispersion (2-lag)	-0.016** (-2.29)	-0.014** (-1.98)	-0.013* (-1.83)	-0.008 (-1.09)	-0.007 (-1.04)	-0.007 (-1.00)	-0.007 (-0.96)	-0.003 (-0.38)	-0.002 (-0.31)
IP Growth (1-lag)	-0.436*** (-30.28)	-0.441*** (-30.28)	-0.442*** (-30.29)	-0.449*** (-29.82)	-0.449*** (-29.81)	-0.449*** (-29.79)	-0.450*** (-29.84)	-0.450*** (-29.82)	-0.449*** (-29.74)
IP Growth (2-lag)	-0.220*** (-15.29)	-0.226*** (-15.49)	-0.226*** (-15.50)	-0.226*** (-15.03)	-0.226*** (-15.00)	-0.226*** (-14.96)	-0.226*** (-15.00)	-0.227*** (-15.00)	-0.226*** (-14.91)
Rate (1-lag)		0.967*** (4.52)	0.947*** (4.42)	0.831*** (3.77)	0.854*** (3.87)	0.833*** (3.76)	0.850*** (3.88)	0.810*** (3.63)	0.854*** (3.82)
Rate (2-lag)		0.168 (0.79)	0.185 (0.87)	0.149 (0.68)	0.111 (0.50)	0.120 (0.54)	0.123 (0.55)	0.218 (0.97)	0.268 (1.18)
MCI (1-lag)			0.009 (1.03)	-0.002 (-0.21)	0.000 (0.01)	0.000 (0.03)	0.002 (0.17)	-0.003 (-0.29)	-0.001 (-0.13)
MCI(2-lag)			0.017* (1.90)	-0.003 (-0.35)	-0.003 (-0.38)	-0.002 (-0.22)	-0.001 (-0.10)	-0.011 (-1.20)	-0.010 (-1.04)
CCI (1-lag)				1.150*** (3.73)	1.162*** (3.77)	1.171*** (3.79)	1.171*** (3.79)	1.118*** (3.58)	1.122*** (3.59)
CCI(2-lag)				0.713** (2.33)	0.703** (2.29)	0.699** (2.28)	0.685** (2.23)	0.639** (2.07)	0.635** (2.06)
EPU - EU (1-lag)					-0.000 (-1.12)	-0.000 (-0.99)	-0.001 (-1.32)	-0.001 (-1.55)	-0.001 (-1.43)
EPU - EU (2-lag)					-0.001* (-1.78)	-0.001 (-1.57)	-0.001 (-1.57)	-0.001 (-1.44)	-0.001 (-1.49)
EPU - US (1-lag)						-0.000 (-0.22)	-0.000 (-0.21)	0.000 (0.59)	0.000 (0.23)
EPU - US (2-lag)						-0.000 (-1.01)	-0.000 (-0.80)	-0.000 (-0.46)	-0.000 (-0.47)
EPU - Japan (1-lag)							0.001** (2.36)	0.001*** (2.63)	0.001*** (2.64)
EPU - Japan (2-lag)							0.001 (1.27)	0.000 (1.09)	0.000 (0.88)
VIX (1-lag)								0.006 (0.59)	0.014 (1.39)
VIX(2-lag)								-0.039*** (-3.97)	-0.040*** (-3.96)
Oil Volatility (1-lag)								-0.085** (-2.48)	-0.085** (-2.48)
Oil Volatility (2-lag)								-0.047 (-1.41)	-0.047 (-1.41)
Constant	0.234*** (7.39)	0.265*** (8.06)	0.265*** (8.05)	0.259*** (7.77)	0.258*** (7.73)	0.258*** (7.72)	0.259*** (7.76)	0.260*** (7.79)	0.262*** (7.84)
Observations	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on IP growth using the two-step [Arellano and Bond \(1991\)](#) estimator with up to 6 lags as instruments. All variables' transformations follow the main texts. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table C.2: Dynamic Regression Results using the Unemployment Rate: With the [Arellano and Bond \(1991\)](#) Estimator

	Dependent Variable: the Unemployment Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	3.185*** (6.70)	2.423*** (5.14)	2.422*** (5.14)	2.380*** (4.93)	2.382*** (4.93)	2.312*** (4.78)	2.313*** (4.77)	2.486*** (5.03)	2.508*** (5.07)
Stock Market Dispersion (2-lag)	2.872*** (5.97)	1.872*** (3.91)	1.909*** (3.98)	1.876*** (3.84)	1.884*** (3.85)	1.876*** (3.82)	1.879*** (3.82)	1.919*** (3.86)	1.929*** (3.88)
Unemployment (1-lag)	0.927*** (61.78)	0.874*** (56.91)	0.873*** (56.83)	0.877*** (55.17)	0.877*** (55.12)	0.877*** (55.13)	0.877*** (55.12)	0.876*** (54.94)	0.875*** (54.85)
Unemployment (2-lag)	0.097*** (6.53)	0.062*** (4.15)	0.064*** (4.24)	0.067*** (4.24)	0.066*** (4.23)	0.067*** (4.24)	0.067*** (4.25)	0.066*** (4.21)	0.066*** (4.17)
Rate (1-lag)		-0.130*** (-10.10)	-0.131*** (-10.00)	-0.132*** (-9.86)	-0.132*** (-9.79)	-0.133*** (-9.80)	-0.133*** (-9.78)	-0.131*** (-9.45)	-0.130*** (-9.26)
Rate (2-lag)		(-9.46)	(-9.55)	(-9.06)	(-9.02)	(-9.08)	(-9.08)	(-8.91)	(-8.80)
		-0.138***	-0.137***	-0.143***	-0.144***	-0.145***	-0.145***	-0.142***	-0.140***
MCI (1-lag)			0.000	0.000	0.000	0.000	0.000	0.000	0.000
MCI(2-lag)			(0.47)	(0.43)	(0.40)	(0.23)	(0.25)	(-0.23)	(-0.19)
			0.001**	0.001*	0.001	0.001*	0.001*	0.001	0.001*
			(2.08)	(1.69)	(1.64)	(1.86)	(1.87)	(1.63)	(1.70)
CCI (1-lag)			0.016	0.016	0.016	0.016	0.016	0.010	0.010
CCI(2-lag)			(0.79)	(0.78)	(0.76)	(0.75)	(0.75)	(0.47)	(0.48)
			0.008	0.009	0.009	0.009	0.009	0.011	0.011
			(0.40)	(0.41)	(0.45)	(0.45)	(0.45)	(0.55)	(0.54)
EPU - EU (1-lag)					-0.000	-0.000	-0.000	-0.000	-0.000
EPU - EU (2-lag)					(-0.43)	(-0.46)	(-0.44)	(-0.45)	(-0.40)
					0.000	0.000	0.000	0.000	0.000
EPU - US (1-lag)					(0.22)	(0.42)	(0.37)	(0.29)	(0.29)
EPU - US (2-lag)						0.000	0.000	0.000	0.000
						(0.97)	(0.97)	(0.97)	(0.81)
EPU - Japan (1-lag)						-0.000	-0.000	-0.000	-0.000
EPU - Japan (2-lag)						(-1.38)	(-1.39)	(-1.52)	(-1.55)
						0.000	0.000	0.000	0.000
						(0.12)	(0.12)	(0.26)	(0.25)
VIX (1-lag)								0.000	0.000
VIX(2-lag)								(0.04)	(0.05)
								-0.001*	-0.001
Oil Volatility (1-lag)								(-1.69)	(-1.24)
Oil Volatility (2-lag)								-0.001	-0.001
								(-0.76)	(-0.73)
								-0.002	-0.002
								(-1.02)	(-1.02)
								-0.002	-0.002
								(-0.81)	(-0.81)
Constant	-0.001 (-0.50)	-0.007*** (-3.08)	-0.007*** (-3.09)	-0.007*** (-2.93)	-0.007*** (-2.93)	-0.007*** (-2.95)	-0.007*** (-2.95)	-0.006*** (-2.88)	-0.006*** (-2.85)
Observations	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688	4,688

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on the unemployment rate using the two-step [Arellano and Bond \(1991\)](#) estimator with up to 6 lags as instruments. All variables' transformations follow the main texts. The unemployment rate is the harmonized unemployment rate from the OECD. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table C.3: Dynamic Regression Results: Unemployment (Log Transformation Robustness)

	Dependent Variable: the Unemployment Rate								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	1.383*** (2.96)	1.061** (2.50)	1.055** (2.53)	1.075** (2.53)	1.079** (2.53)	1.046** (2.38)	1.054** (2.38)	1.117** (2.31)	1.110** (2.30)
Stock Market Dispersion (2-lag)	0.852** (2.18)	0.338 (1.13)	0.358 (1.18)	0.413 (1.29)	0.407 (1.28)	0.409 (1.25)	0.423 (1.28)	0.367 (1.18)	0.373 (1.18)
Unemployment (1-lag)	0.984*** (6.89)	0.960*** (6.63)	0.960*** (6.62)	0.960*** (6.33)	0.960*** (6.33)	0.960*** (6.34)	0.960*** (6.33)	0.960*** (6.32)	0.960*** (6.33)
Unemployment (2-lag)	-0.070 (-0.51)	-0.049 (-0.35)	-0.050 (-0.36)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)	-0.051 (-0.35)
Rate (1-lag)	-0.072*** (-6.93)	-0.072*** (-6.93)	-0.073*** (-7.04)	-0.074*** (-6.70)	-0.075*** (-6.75)	-0.075*** (-6.83)	-0.075*** (-6.85)	-0.075*** (-6.70)	-0.075*** (-6.75)
Rate (2-lag)	-0.069** (-2.58)	-0.069** (-2.58)	-0.068** (-2.56)	-0.076** (-2.78)	-0.075** (-2.79)	-0.076** (-2.71)	-0.075** (-2.70)	-0.075** (-2.69)	-0.075** (-2.61)
Log MCI (1-lag)			-0.012 (-0.46)	-0.004 (-0.13)	-0.006 (-0.19)	-0.014 (-0.39)	-0.010 (-0.27)	-0.025 (-0.71)	-0.026 (-0.73)
Log MCI (2-lag)			0.048 (1.46)	0.045 (1.19)	0.048 (1.23)	0.054 (1.41)	0.057 (1.47)	0.060 (1.64)	0.060 (1.67)
Log CCI (1-lag)				1.027 (0.39)	1.005 (0.38)	0.991 (0.38)	0.997 (0.38)	0.699 (0.27)	0.708 (0.27)
Log CCI (2-lag)				-0.262 (-0.13)	-0.265 (-0.13)	-0.249 (-0.12)	-0.283 (-0.14)	-0.092 (-0.04)	-0.092 (-0.04)
Log EPU - EU (1-lag)					0.002 (1.08)	0.001 (0.90)	0.001 (0.82)	0.002 (0.98)	0.002 (0.93)
Log EPU - EU (2-lag)					0.002 (0.67)	0.003 (0.78)	0.002 (0.64)	0.002 (0.58)	0.002 (0.62)
Log EPU - US (1-lag)						0.002 (0.68)	0.002 (0.66)	0.002 (0.53)	0.002 (0.52)
Log EPU - US (2-lag)						-0.002 (-0.82)	-0.002 (-0.79)	-0.002 (-0.85)	-0.002 (-0.88)
Log EPU - Japan (1-lag)							0.001 (0.01)	0.001 (0.53)	0.001 (0.53)
Log EPU - Japan (2-lag)							0.002* (2.03)	0.002* (1.87)	0.002* (1.92)
Δ Log VIX (1-lag)								-0.024 (-1.47)	-0.024 (-1.42)
Δ Log VIX (2-lag)								0.004 (0.30)	0.005 (0.34)
Log Oil Volatility (1-lag)									0.025 (0.27)
Log Oil Volatility (2-lag)									-0.023 (-0.25)
Constant	-0.019*** (-3.10)	-0.015*** (-3.01)	-0.015*** (-3.00)	-0.016*** (-3.06)	-0.016*** (-3.06)	-0.016*** (-3.04)	-0.016*** (-3.03)	-0.016*** (-2.98)	-0.016*** (-2.97)
Observations	4,448	4,333	4,333	4,050	4,050	4,050	4,050	4,050	4,050
Adj - R^2	0.848	0.852	0.852	0.854	0.854	0.854	0.854	0.854	0.854

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on the unemployment rate. The interest rate (Rate), MCI, CCI, oil volatility, the VIX, and all the EPU's are first-differenced. Stock market dispersion and its lags are not first-differenced. Sample periods: 1994M1-2017M5. t statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

Table C.4: Dynamic Regression Results: IP Growth (Log Transformation Robustness)

	Dependent Variable: Industrial Production Growth								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Stock Market Dispersion (1-lag)	-0.035** (-2.51)	-0.032** (-2.31)	-0.031** (-2.28)	-0.028* (-2.06)	-0.028* (-2.07)	-0.028* (-2.07)	-0.028* (-2.06)	-0.029* (-1.95)	-0.028* (-1.94)
Stock Market Dispersion (2-lag)	-0.023*** (-5.46)	-0.017*** (-5.13)	-0.017*** (-5.07)	-0.015*** (-4.58)	-0.015*** (-4.62)	-0.015*** (-4.68)	-0.014*** (-4.56)	-0.013*** (-3.57)	-0.013*** (-3.49)
IP Growth (1-lag)	-0.326*** (-7.06)	-0.340*** (-8.08)	-0.341*** (-8.18)	-0.341*** (-7.60)	-0.341*** (-7.62)	-0.341*** (-7.61)	-0.341*** (-7.67)	-0.341*** (-7.67)	-0.341*** (-7.58)
IP Growth (2-lag)	-0.133*** (-3.52)	-0.148*** (-4.65)	-0.149*** (-4.67)	-0.145*** (-4.31)	-0.145*** (-4.29)	-0.145*** (-4.28)	-0.146*** (-4.32)	-0.147*** (-4.28)	-0.145*** (-4.10)
Rate (1-lag)		1.148*** (5.41)	1.132*** (5.38)	1.140*** (6.13)	1.147*** (6.00)	1.140*** (5.85)	1.146*** (5.94)	1.133*** (5.76)	1.104*** (5.62)
Rate (2-lag)		0.352* (1.90)	0.384** (2.14)	0.410** (2.16)	0.404** (2.27)	0.409** (2.20)	0.404** (2.19)	0.414** (2.27)	0.489** (2.65)
Log MCI (1-lag)			0.788* (1.80)	0.199 (0.41)	0.281 (0.60)	0.258 (0.55)	0.342 (0.73)	0.428 (0.93)	0.406 (0.88)
Log MCI (2-lag)			1.319* (1.95)	-0.091 (-0.14)	-0.124 (-0.19)	-0.052 (-0.08)	0.064 (0.10)	-0.202 (-0.35)	-0.083 (-0.15)
Log CCI (1-lag)				46.321 (1.72)	47.261 (1.75)	47.479* (1.77)	48.483* (1.83)	51.628* (2.09)	47.819* (1.95)
Log CCI (2-lag)				59.900*** (2.99)	59.037*** (3.00)	58.896*** (3.01)	56.934*** (3.00)	53.539** (2.91)	53.580** (2.96)
Log EPU - EU (1-lag)					-0.013 (-0.29)	-0.010 (-0.24)	-0.028 (-0.76)	-0.033 (-1.01)	-0.023 (-0.67)
Log EPU - EU (2-lag)					-0.064** (-2.57)	-0.058** (-2.41)	-0.060** (-2.48)	-0.052** (-2.21)	-0.052** (-2.30)
Log EPU - US (1-lag)						-0.002 (-0.04)	0.000 (0.01)	0.010 (0.29)	-0.012 (-0.35)
Log EPU - US (2-lag)						-0.027 (-0.62)	-0.020 (-0.48)	-0.020 (-0.74)	-0.031 (-0.74)
Log EPU - Japan (1-lag)							0.071** (2.48)	0.082** (2.35)	0.065* (1.93)
Log EPU - Japan (2-lag)							0.025 (1.58)	0.026 (1.52)	0.014 (0.84)
Δ Log VIX (1-lag)								0.355 (1.13)	0.482 (1.13)
Δ Log VIX(2-lag)								-0.372* (-1.77)	-0.327 (-1.54)
Log Oil Volatility (1-lag)									-6.441*** (-3.87)
Log Oil Volatility (2-lag)									-4.487*** (-3.18)
Constant	0.228*** (16.92)	0.268*** (19.61)	0.268*** (19.79)	0.270*** (17.52)	0.270*** (17.32)	0.270*** (17.31)	0.270*** (17.29)	0.271*** (17.47)	0.271*** (17.16)
Observations	4,368	4,269	4,269	3,990	3,990	3,990	3,990	3,990	3,990
Adj R ²	0.105	0.117	0.118	0.121	0.121	0.121	0.121	0.122	0.124

Note: This table presents the results from a set of dynamic panel regressions of stock market dispersion on IP growth. The interest rate (Rate), MCI, CCI, oil volatility, the VIX, and all the EPUs are first-differenced. Stock market dispersion and its lags are not first-differenced. Sample periods: 1994M1-2017M5. *t* statistics in parentheses. * denotes $p < 0.10$, ** denotes $p < 0.05$, *** denotes $p < 0.01$

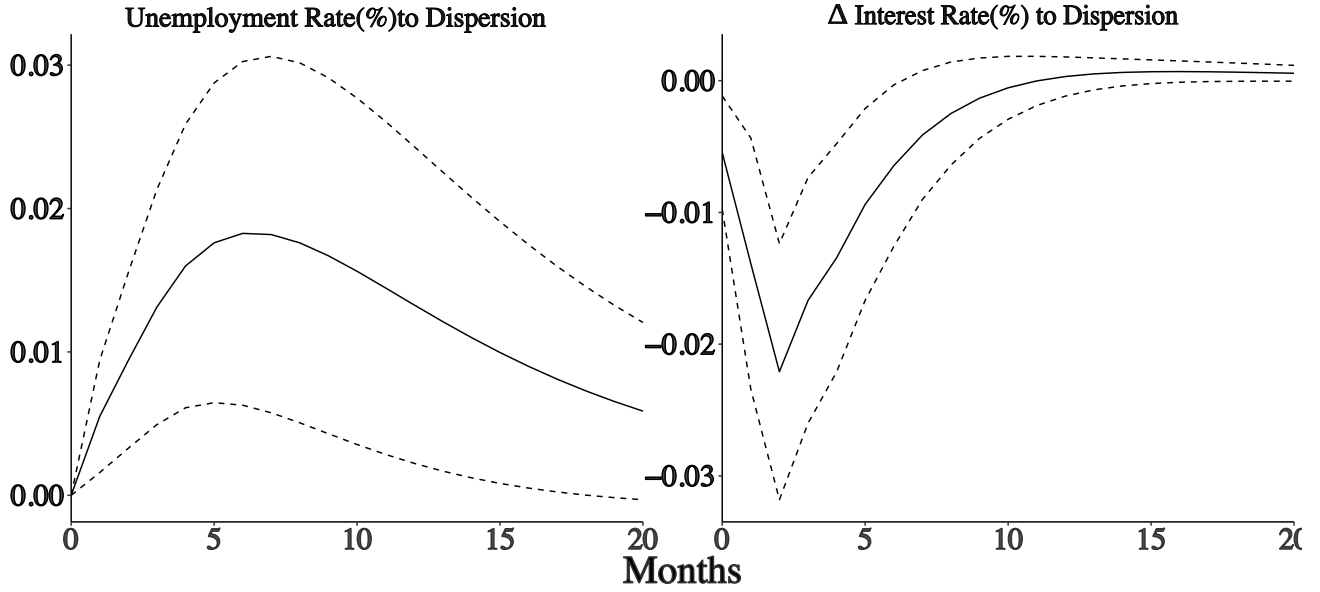
Table C.5: Dynamic Panel Regressions with Labor Market Efficiency Index

Dependent Variable:	IP Growth	Unemployment
	(1)	(2)
Dispersion (1-lag)	-0.020 (-1.25)	1.726 (1.27)
Dispersion (2-lag)	-0.019** (-2.40)	0.848 (0.93)
Labor Market Efficiency Index (LME) (1-lag)	3.020 (0.89)	-0.203* (-2.00)
Labor Market Efficiency Index (LME) (2-lag)	4.353 (1.25)	-0.323** (-2.40)
Interaction (LME x Dispersion) (1-lag)	0.044 (0.33)	23.082* (1.84)
Interaction (LME x Dispersion) (2-lag)	-0.401* (-2.02)	27.909** (2.40)
Dependent Variable (1-lag)	-0.260*** (-5.45)	0.979*** (6.18)
Dependent Variable (2-lag)	-0.173** (-2.84)	-0.073 (-0.51)
Rate (1-lag)	2.017*** (3.89)	-0.095** (-2.32)
Rate (2-lag)	0.533* (2.06)	-0.143** (-2.91)
MCI (1-lag)	0.015 (0.96)	-0.001 (-1.21)
MCI (2-lag)	-0.020 (-1.65)	0.000 (0.45)
CCI (1-lag)	0.402 (1.11)	-0.002 (-0.05)
CCI (2-lag)	0.427 (1.02)	0.010 (1.11)
Constant	0.126*** (8.15)	-0.031 (-1.68)
Observations	1,755	1,815
Adj – R^2	0.119	0.893

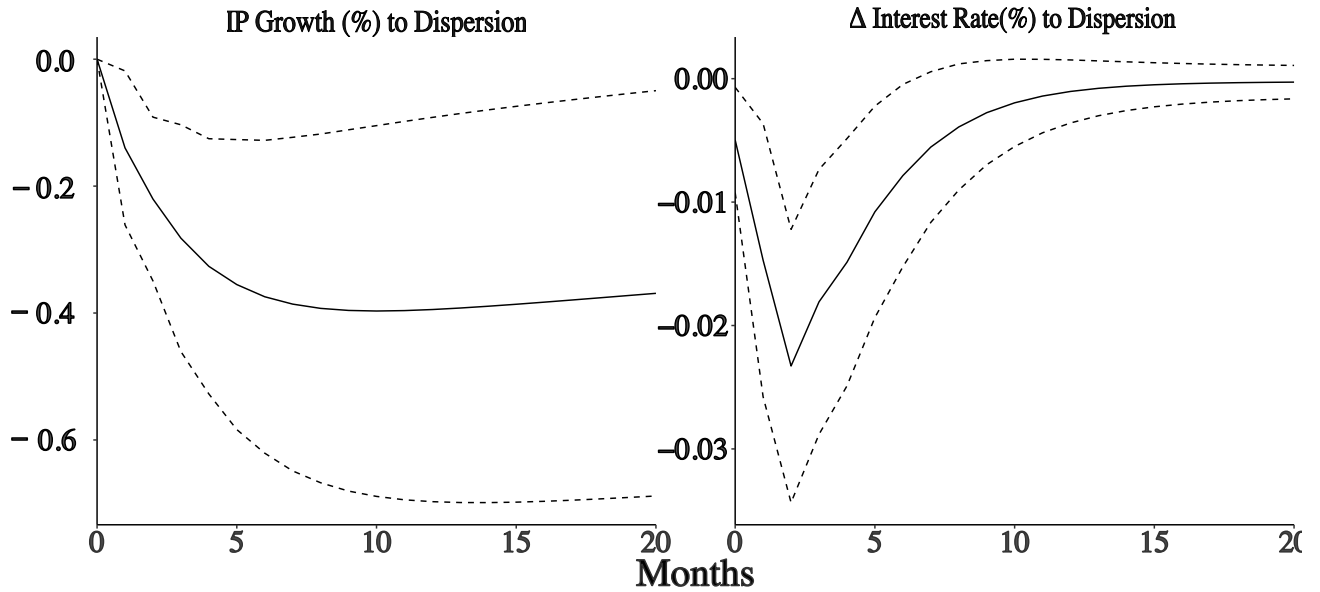
Note: The interaction terms are between (i) stock market dispersion and (ii) the global market index. Controls that are included but not reported: EPU's for the EU, Japan, and the U.S., the CBEO:VIX, and Oil Volatility. t statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure C.1: Baseline IRF: Alternative Ordering

Panel VAR Order: $[Unemp, Dispersion, \Delta Rate]$



Panel VAR Order: $[IP, Dispersion, \Delta Rate]$



Note: This figure presents the impulse responses of IP growth (in percentage) and the interest rate (in percentage) to a positive one-standard-deviation shock to stock market dispersion with an alternative ordering. Confidence bands (95%) are generated using 2,000 replications.